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From Cosmos to Clinic: Interpretable Spatial Statistics for Histopathology Prognosis

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Abstract

Current interpretability methods in computational pathology like attention heatmaps, identify what tissue morphologies are predictive, but not how their spatial interactions affect clinical outcomes. To bridge this gap, we introduce a novel, handcrafted spatial representation for whole slide images (WSIs) based on the Landy-Szalay estimator corrected two-point correlation function (2PCF), adapted from astrophysics. Our method quantifies complex spatial relationships between morphological classes and translates them into natural language sentences describing prognostic spatial relations (e.g., "Morphology A interspersed with Morphology B is associated with poor outcome" or "Morphology A at a distance of "x" μm from Morphology B is associated with good outcome"). We show that our interpretable spatial features capture prognostic signals orthogonal to those used in Multiple Instance Learning (MIL), and their fusion significantly surpasses MIL baselines on ovarian cancer response prediction, exceeding the predictive value of current clinical genetic tests. By surfacing spatial biomarkers in human-readable form, our approach offers a new tool for biological insight into treatment resistance.

1. Introduction

The field of computational pathology has achieved remarkable success in developing predictive models for clinical outcomes, with several approaches nearing FDA approval for tasks like Gleason grading [2, 15] and cancer detection [3, 17]. Multiple Instance Learning (MIL) has emerged as the dominant modeling paradigm, with attention-based methods [10, 13], advances like TransMIL [16], and foundation model approaches such as TITAN [7] achieving impressive performance across diverse tasks. However, these successes have been primarily measured by predictive accuracy rather than biological interpretability.

1.1. The Interpretability Gap in Spatial Understanding

Despite their clinical promise, current computational pathology models suffer from a fundamental limitation: they provide minimal insight into the spatial biological processes driving their predictions. While attention mechanisms can generate heatmaps highlighting "important" regions [10, 16], these visualizations offer no understanding of **how** different tissue morphologies spatially interact to influence outcomes. Pathologists, in contrast, rely heavily on spatial architectural patterns—immune cell infiltration patterns [8, 20], tumor-stroma boundary characteristics [1], stromal organization, and cellular spatial distributions [19]—to make diagnostic and prognostic assessments. Current heatmap-based interpretability methods are simply not designed to capture these spatial relationships, limiting our ability to extract biological knowledge from computational models. While recent advances have made significant strides—PAMIL's prototype-based attention mechanisms enhance interpretability through case-based reasoning [12], xMIL's layer-wise relevance propagation provides positive/negative evidence distinction [9], and PathChat demonstrates sophisticated multimodal AI assistance [14]—these approaches remain fundamentally limited to identifying "where the model looks" rather than quantifying "how spatial tissue organization influences outcome." They focus on instance-level explanations, attention-based visualizations, or conversational interfaces, but lack systematic methods for characterizing and describing the spatial architectural patterns that pathologists rely upon for prognostic assessment.

1.2. From Prediction to Biological Insight

Rather than pursuing marginal improvements in predictive accuracy, we reframe computational pathology as a **biological knowledge discovery** problem. Our goal is not to achieve state-of-the-art performance on digital pathology benchmarks but to develop methods that can automatically uncover interpretable spatial patterns in tissue organization.

We introduce what we term **"spatial interpretability"**: the ability to generate natural language descriptions of spatial relationships between tissue morphologies that directly relate to biological processes. Unlike traditional attention-based interpretability, spatial interpretability provides quantitative, scale-specific insights into tissue architecture that can be validated by pathologists and translated into biological hypotheses.

1.3. Technical Approach and Contributions

To achieve spatial interpretability, we adapt the two-point correlation function (2PCF) from astrophysics [11]—a tool originally designed to quantify galaxy clustering across cosmic scales—to characterize tissue architecture at microscopic scales. While astronomers use 2PCF to probe the cosmic web on megaparsec scales, we apply the same mathematical principles to quantify tumor microenvironment organization on micrometer scales.

A critical technical challenge is that whole-slide images represent 2D slices of 3D tumor tissue, creating irregular boundaries with complex edge effects that bias spatial measurements. Without proper correction, these geometric artifacts would completely obscure true biological spatial relationships, rendering correlation-based analysis meaningless. We address this through the Landy-Szalay estimator [11], which provides robust correction for these geometric artifacts—effectively unlocking reliable spatial analysis in computational pathology for the first time. This correction is essential not just for accuracy, but for the very feasibility of extracting interpretable spatial biomarkers from tissue sections. Our complete workflow (Figure 1) demonstrates this integrated approach from raw histology to interpretable spatial biomarkers. Our contributions are threefold:

1. **Novel Spatial Representation**: We introduce a complete pipeline for extracting handcrafted spatial features from WSIs using 2PCF with Landy-Szalay edge correction, providing the first method capable of generating natural language descriptions of tissue spatial relationships.
2. **Biological Knowledge Discovery**: We demonstrate that our interpretable spatial features capture orthogonal prognostic signals to MIL, revealing specific spatial biomarkers like "homogeneous tumor regions predict poor platinum response" that provide direct biological insights into treatment mechanisms.
3. **Methodological Framework**: We establish a transparent, invertible modeling framework that enables systematic exploration of spatial architecture in cancer, offering a new paradigm for biological discovery in computational pathology.

Our work demonstrates that handcrafted representations, when grounded in established statistical principles and domain knowledge, can provide unique biological insights that complement rather than simply improve upon deep learning approaches.

2. Mathematical Framework: Handcrafted Spatial Representation

Our method rigorously quantifies tissue spatial organization through statistical correlation analysis, transforming whole-slide images into interpretable features while preserving biological meaning at each computational step.

2.1. Morphological Discretization and Spatial Point Process Formation

Patch Extraction and High-Dimensional Embedding: Let $\mathcal{I}$ denote a gigapixel whole-slide image (Figure 1a) scanned at 20× magnification with tissue region $W \subset \mathbb{R}^2$. We extract non-overlapping patches P_k of size 256×256 pixels from tissue regions, yielding a set of spatial coordinates $\{(x_k, y_k)\}_{k=1}^{N}$ where N is the total number of patches for the slide.

Each patch P_k is processed through a foundation model encoder $\phi : \mathbb{R}^{256\times256\times3} \to \mathbb{R}^d$ (e.g., UNI [5]) to produce high-dimensional feature embeddings $\mathbf{h}_k = \phi(P_k) \in \mathbb{R}^d$ where $d = 1024$.

Cohort-Level Morphological Prototype Discovery: Given the complete training cohort $\mathcal{D} = \{\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_M\}$, we pool all patch embeddings $\mathbf{H} = \{\mathbf{h}_{i,j}\}$ where $i \in [1, M]$ indexes slides and $j \in [1, N_i]$ indexes patches within slide i.

Following the morphological prototype discovery approach introduced in PANTHER [4], we apply K-means clustering to discover $K = 16$ morphological prototypes (Figure 1b):

$$\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_K\} = \arg\min_{\{\boldsymbol{\mu}_k\}} \sum_{i=1}^{M} \sum_{j=1}^{N_i} \min_{k\in[1,K]} \|\mathbf{h}_{i,j} - \boldsymbol{\mu}_k\|_2^2 \tag{1}$$

Each patch is assigned to its nearest prototype via $c_{i,j} = \arg\min_k \|\mathbf{h}_{i,j} - \boldsymbol{\mu}_k\|_2$, creating a discrete spatial point process where each location $(x_{i,j}, y_{i,j})$ is labeled with morphology prototype $c_{i,j} \in \{1, 2, \ldots, K\}$.

Pathologist-Guided Semantic Mapping: The prototype centers $\{\boldsymbol{\mu}_k\}_{k=1}^{K}$ are annotated by a senior pathologist, establishing a bijection $\mathcal{T} : \{1, 2, \ldots, K\} \to \mathcal{L}$ where $\mathcal{L}$

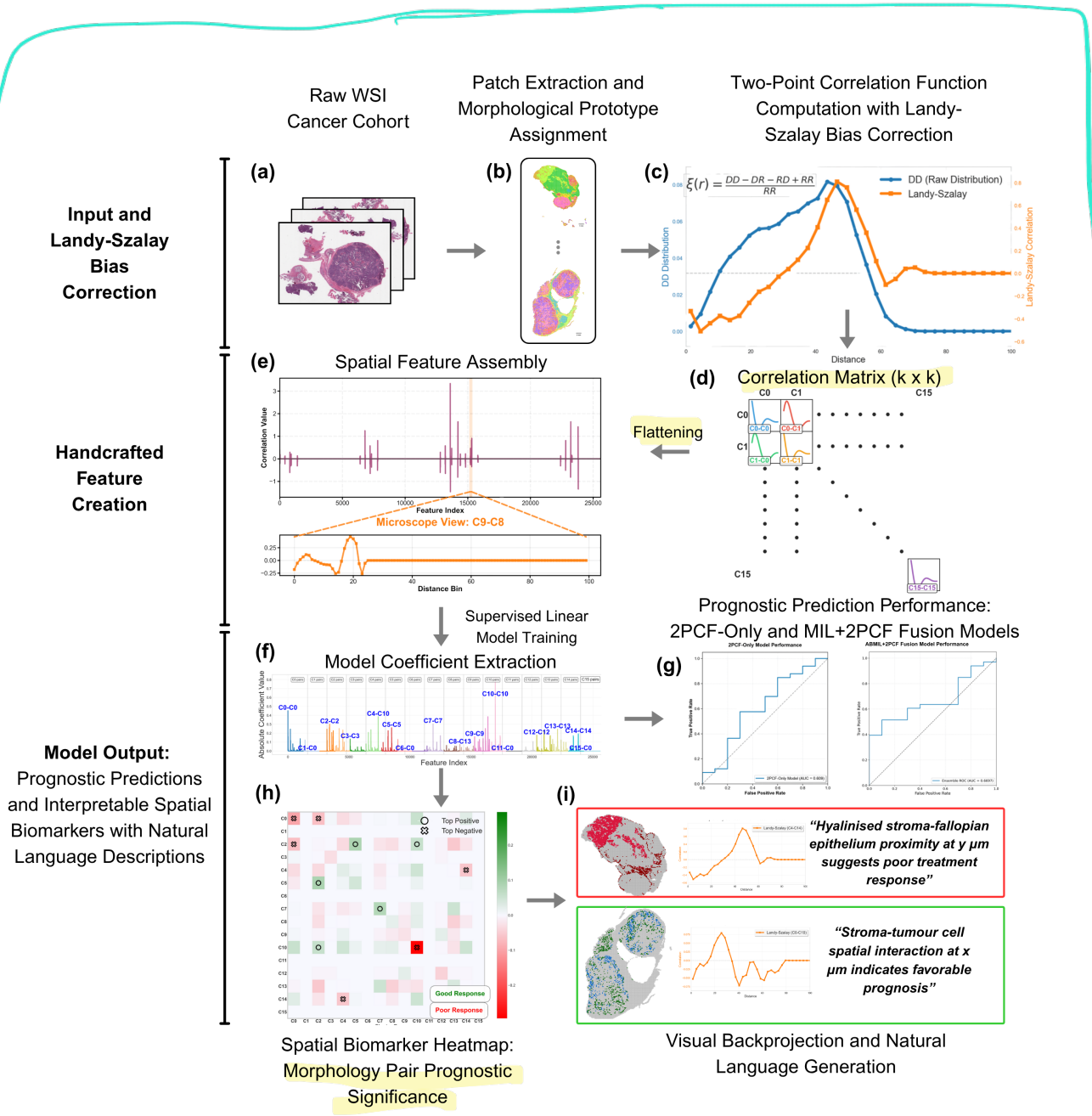


Figure 1. Complete spatial biomarker discovery workflow for computational pathology. (a) Raw whole-slide image from ovarian cancer cohort. (b) Patch extraction and morphological prototype assignment (k=16 clusters) with pathologist annotation of 16 tissue types. (c) Two-point correlation function computation with Landy-Szalay bias correction to remove edge effects inherent in irregular tissue boundaries. (d) Multi-scale correlation matrix assembly capturing spatial relationships between all morphology pairs across distance bins. (e) Feature vector flattening and spatial feature creation, transforming correlation matrices into high-dimensional interpretable features. (f) Supervised linear ML model training and coefficient extraction revealing prognostically significant spatial patterns. (g) ROC curve analysis demonstrating performance of 2PCF-only and MIL+2PCF fusion approaches on platinum response prediction. (h) Spatial biomarker heatmap showing prognostic significance of morphology pair interactions, with green indicating favorable prognosis patterns and red indicating poor prognosis patterns. (i) Visual backprojection of key biomarkers onto tissue and natural language generation producing interpretable statements. This framework enables automatic biological knowledge discovery through explicit spatial modeling while maintaining full transparency and interpretability.

represents established pathological terminology (e.g., "tumor cells," "necrosis," "stroma").

2.2. Two-Point Correlation Function: Rigorous Statistical Foundation

Spatial Point Process Formulation: For each WSI, we now have K distinct point processes $\mathcal{P}_i = \{(x_k, y_k) : c_k = i\}$ for $i \in [1, K]$, each representing the spatial distribution of morphology type i within the irregular observation window W.

Formal Definition of the Cross-Correlation Function: Let ρ_i and ρ_j denote the mean number densities of morphology types i and j within domain W. The cross-correlation function $\xi_{ij}(r)$ between types i and j is defined through the joint probability relation:

$$dP_{ij} = \rho_i \rho_j [1 + \xi_{ij}(r)]\, dV_1\, dV_2 \tag{2}$$

where dP_{ij} represents the probability of finding a patch of type i in infinitesimal volume dV_1 around position $\mathbf{r}_1$ and a patch of type j in dV_2 around position $\mathbf{r}_2$, with $|\mathbf{r}_1 - \mathbf{r}_2| = r$.

For a completely spatially random (Poisson) distribution, $\xi_{ij}(r) = 0$ for all r. Deviations from zero quantify spatial organization:

- $\xi_{ij}(r) > 0$: Excess clustering of types i and j at separation r
- $\xi_{ij}(r) < 0$: Spatial avoidance or anti-correlation at separation r

Discretized Pair-Counting Estimators: To estimate $\xi_{ij}(r)$ from finite point sets, we partition the distance domain into bins $[r_b, r_b + \Delta r]$ for $b \in [1, B]$ where $B = 100$, and employ indicator functions $\mathbf{1}_{[r_b, r_b+\Delta r]}(\cdot)$ to count pairs.

For morphology types i and j, we define the fundamental pair counts:

Data-Data Pairs:

$$DD_{ij}(r_b) = \sum_{\mathbf{p} \in \mathcal{P}_i} \sum_{\mathbf{q} \in \mathcal{P}_j} \mathbf{1}_{[r_b, r_b+\Delta r]}(|\mathbf{p} - \mathbf{q}|) \cdot \mathbf{1}_{i \neq j \text{ or } \mathbf{p} \neq \mathbf{q}} \tag{3}$$

Random-Random Pairs:

$$RR_{ij}(r_b) = \sum_{\mathbf{u} \in \mathcal{R}_i} \sum_{\mathbf{v} \in \mathcal{R}_j} \mathbf{1}_{[r_b, r_b+\Delta r]}(|\mathbf{u} - \mathbf{v}|) \cdot \mathbf{1}_{\mathbf{u} \neq \mathbf{v}} \tag{4}$$

where $\mathcal{R}_i$ and $\mathcal{R}_j$ are random catalogs with $|\mathcal{R}_i| = |\mathcal{P}_i|$ and $|\mathcal{R}_j| = |\mathcal{P}_j|$, uniformly distributed within the same observation window W.

Cross-Correlation Pairs:

$$DR_{ij}(r_b) = \sum_{\mathbf{p} \in \mathcal{P}_i} \sum_{\mathbf{v} \in \mathcal{R}_j} \mathbf{1}_{[r_b, r_b+\Delta r]}(|\mathbf{p} - \mathbf{v}|) \tag{5}$$

$$RD_{ij}(r_b) = \sum_{\mathbf{u} \in \mathcal{R}_i} \sum_{\mathbf{q} \in \mathcal{P}_j} \mathbf{1}_{[r_b, r_b+\Delta r]}(|\mathbf{u} - \mathbf{q}|) \tag{6}$$

2.3. Landy-Szalay Estimator: Rigorous Edge Correction

The Boundary Bias Problem: In finite observation windows with irregular boundaries (characteristic of tissue sections), points near edges suffer from incomplete sampling neighborhoods. Let $\gamma_W(\mathbf{r})$ denote the set-covariance function:

$$\gamma_W(\mathbf{r}) = \int_W \mathbf{1}_W(\mathbf{s}) \mathbf{1}_W(\mathbf{s} + \mathbf{r})\, d\mathbf{s} \tag{7}$$

This geometric factor quantifies the overlap between domain W and its translation by vector $\mathbf{r}$, directly affecting pair count expectations near boundaries.

Landy-Szalay Estimator Derivation: Following Landy & Szalay [11], the unbiased estimator for $\xi_{ij}(r)$ accounts for boundary effects through (Figure 1c):

$$\hat{\xi}_{ij}^{LS}(r_b) = \frac{DD_{ij}(r_b) - DR_{ij}(r_b) - RD_{ij}(r_b) + RR_{ij}(r_b)}{RR_{ij}(r_b)} \tag{8}$$

Theoretical Justification: The estimator's bias correction arises from the expectation relationships:

$$\mathbb{E}[DD_{ij}(r_b)] = \rho_i \rho_j [1 + \xi_{ij}(r_b)] \gamma_W(r_b) \Delta r \tag{9}$$

$$\mathbb{E}[DR_{ij}(r_b)] = \mathbb{E}[RD_{ij}(r_b)] = \rho_i \rho_j \gamma_W(r_b) \Delta r \tag{10}$$

$$\mathbb{E}[RR_{ij}(r_b)] = \rho_i \rho_j \gamma_W(r_b) \Delta r \tag{11}$$

The geometric factor $\gamma_W(r_b)$ cancels in the ratio, yielding:

$$\mathbb{E}[\hat{\xi}_{ij}^{LS}(r_b)] = \xi_{ij}(r_b) \tag{12}$$

ensuring unbiased estimation regardless of boundary complexity.

2.4. Multi-Scale Feature Tensor Construction

Correlation Tensor Formation: For each WSI, we construct the complete correlation tensor (Figure 1d):

$$\mathbf{\Xi} \in \mathbb{R}^{K \times K \times B} \tag{13}$$

16 × 16 × 100 = 25600

where $\mathbf{\Xi}_{ij,b} = \hat{\xi}_{ij}^{LS}(r_b)$ represents the spatial correlation between morphology types i and j at distance scale r_b.

Vectorization and Feature Space: The tensor is flattened into a high-dimensional feature vector as a WSI representation (Figure 1e):

$$\mathbf{f} = \text{vec}(\mathbf{\Xi}) \in \mathbb{R}^{K^2 \cdot B} = \mathbb{R}^{25600} \tag{14}$$

where each dimension $f_{(i-1)K+j,b}$ corresponds to the spatial relationship between morphology pair (i, j) at distance bin b.

Dimensionality Reduction via Principal Component Analysis: To address potential overfitting and extract dominant spatial patterns, we apply PCA:

$$\mathbf{f}_{PCA} = \mathbf{U}^T (\mathbf{f} - \boldsymbol{\mu}_f) \tag{15}$$

where $\mathbf{U} \in \mathbb{R}^{25600\times100}$ contains the top 100 principal components and $\boldsymbol{\mu}_f$ is the feature mean across the training cohort.

Linear Classification with Interpretable Weights: The reduced features are classified using a linear SVM (Figure 1f):

$$\hat{y} = \text{sign}(\mathbf{w}^T \mathbf{f}_{PCA} + b) \tag{16}$$

The linear formulation enables complete model interpretability through back-projection:

$$\boldsymbol{\beta}_{original} = \mathbf{U}\mathbf{w} \tag{17}$$

where $\boldsymbol{\beta}_{original} \in \mathbb{R}^{25600}$ provides direct weights for each morphology-pair-distance combination, enabling identification of prognostically significant spatial patterns.

2.5. Spatial Biomarker Identification and Tissue Backprojection

From the back-projected coefficient vector $\boldsymbol{\beta}_{original} \in \mathbb{R}^{25600}$, we identify the most prognostically significant morphology pairs using the weighted mean metric. For each pair (i, j), we extract the corresponding 100-dimensional coefficient sub-vector $\boldsymbol{\beta}_{ij} = [\beta_{ij,1}, \beta_{ij,2}, \ldots, \beta_{ij,100}]$ and compute:

$$\text{importance}_{ij} = \frac{\sum_{b=1}^{B} \beta_{ij,b} \cdot |\beta_{ij,b}|}{\sum_{b=1}^{B} |\beta_{ij,b}|} \tag{18}$$

Pairs are ranked by $|\text{importance}_{ij}|$ to identify the top-k most prognostically relevant spatial interactions.

For each highly-ranked pair (i, j), we reconstruct the complete two-point correlation function by mapping the coefficient weights back to the original 2PCF values:

$$\xi^*_{ij}(r_b) = \Xi_{ij,b} \cdot \text{sign}(\beta_{ij,b}) \tag{19}$$

where $\Xi_{ij,b}$ represents the original Landy-Szalay corrected correlation values and the sign indicates the prognostic direction learned by the model.

Each reconstructed correlation function $\xi^*_{ij}(r)$ is matched against our predefined taxonomy of spatial relationship patterns based on curve characteristics such as peak positions, decay rates, magnitude, and overall functional form. This classification system maps specific 2PCF curve shapes to standardized spatial relationship descriptors.

Finally, we backproject the identified prototype pairs onto whole-slide images by highlighting all tissue regions where morphology types i and j exhibit the characteristic spatial relationship captured by $\xi^*_{ij}(r)$. This produces interpretable spatial biomarkers with direct visual correspondence to specific tissue regions, enabling pathologists to validate computational findings against morphological observations and generating natural language descriptions of the spatial patterns driving prognostic predictions.

Table 1. Pathologist-annotated labels for the 16 tissue clusters.

ID	Pathologist Label	ID	Pathologist Label
0	Stroma and tumour cells	8	Tumour cells + stroma
1	Stroma with capillaries	9	Stroma
2	Desquamated tumour cells	10	Tumour cells
3	Tumour cells	11	Tumour cells
4	Tumour cells + hyalinised stroma	12	Tumour cells
5	Stroma	13	Old haemorrhage
6	Tumour cells	14	Normal fallopian-tube epithelium
7	Necrosis	15	Peri-tubal connective-tissue stroma

Table 2. Performance comparison on platinum response prediction.

Model	Test AUC
Attention-MIL (Baseline)	0.59
2PCF Spatial Features (Ours)	0.61
Fusion Model (MIL + Spatial)	**0.67**

3. Experiments and Results

3.1. Experimental Setup

Dataset and Clinical Task: We evaluate on high-grade serous ovarian carcinoma (HGSOC) platinum response prediction, chosen because: (1) it represents a challenging clinical problem with 20-25% non-response rates [6], (2) spatial microenvironment organization is known to influence treatment efficacy [20], and (3) the inherent difficulty demonstrates our method's ability to extract signal from noisy slide-level labels.

The dataset comprises 242 training cases from a proteogenomic study [6] and an independent test cohort from TCGA-OV [18]. This challenging task has proven difficult for computational approaches.

Morphology Annotation: Table 1 shows the 16 pathologist-annotated morphology clusters used as input to our spatial analysis pipeline.

Baseline Comparison: We compare against attention-based MIL [10] using identical train-test splits and 5-fold cross-validation. Final test performance uses ensemble predictions from all folds.

3.2. Quantitative Results

Table 2 presents our main results. Our handcrafted spatial features achieve test AUC of 0.61 (Figure 1g), outperforming the attention-MIL baseline (0.59) despite using only spatial organization information and a simple linear classifier.

Orthogonal Signal Capture: The fusion model achieving 0.67 AUC demonstrates that spatial features capture prognostic information orthogonal to MIL's learned representations, supporting our hypothesis that handcrafted spatial analysis provides complementary biological insights.

3.3. Spatial Interpretability: Biological Knowledge Discovery

Model Inversion for Spatial Understanding: The linear framework enables complete model inversion. We back-project SVM weights from PCA space to the original 25,600-dimensional spatial feature space, revealing which specific morphology interactions at which spatial scales drive predictions. The spatial biomarker heatmap (Figure 1h) reveals key morphology interactions, while visual backprojection (Figure 1i) enables direct pathological validation.

Spatial Biomarker Discovery: Analysis of the learned weights reveals key biological insights:

1. **Homogeneous Tumor Clustering (C10-C10)**: Strong negative weights indicate that large, spatially homogeneous tumor regions predict poor platinum response—consistent with biological understanding of hypoxic, drug-resistant tumor cores.
2. **Necrotic Tissue Organization (C7-C7)**: Positive weights for necrotic tissue clustering associate with better outcomes, potentially reflecting effective immune-mediated tumor destruction.
3. **Tumor-Stroma Interface Patterns**: Multiple off-diagonal interactions highlight the critical role of tumor microenvironment spatial organization in treatment response.

Natural Language Spatial Descriptions: These findings translate directly into interpretable statements about tissue architecture:

- "Spatially clustered tumor cells at 50-150 μm scales predict poor platinum response"
- "Organized necrotic regions with defined boundaries associate with better treatment outcomes"
- "Heterogeneous tumor-stroma interfaces correlate with improved drug response"

3.4. Landy-Szalay Correction: Unlocking Spatial Analysis

The Landy-Szalay correction proves essential not merely for improved accuracy, but for enabling meaningful spatial analysis altogether. Comparing corrected vs. uncorrected features demonstrates a dramatic performance difference (AUC 0.609 vs. 0.548), but more critically, the uncorrected approach produces spatial correlation measurements that are dominated by boundary artifacts rather than biological organization.

Without edge correction, the extracted spatial features primarily reflect tissue section geometry—irregular boundaries, cutting artifacts, and sampling biases—rather than true tumor microenvironment architecture. The Landy-Szalay estimator effectively removes these confounding factors, revealing the underlying biological spatial relationships that drive our interpretable biomarker discovery.

4. Discussion

4.1. Biological Knowledge Discovery vs. Predictive Performance

Our work demonstrates that the pursuit of biological understanding can yield both interpretable insights and competitive predictive performance. Rather than optimizing for marginal AUC improvements on inherently noisy clinical endpoints, we have developed a framework that reveals interpretable spatial biomarkers with direct biological meaning.

The success of our handcrafted approach challenges the assumption that end-to-end deep learning is always superior for medical imaging tasks. By incorporating domain knowledge through statistical principles established in astrophysics, we achieve spatial interpretability impossible with purely data-driven approaches.

4.2. Spatial Interpretability: Beyond Attention Heatmaps

Traditional attention-based interpretability provides limited insight into the spatial biological processes driving predictions. Our spatial interpretability framework generates natural language descriptions of tissue relationships that can be directly validated by pathologists and translated into biological hypotheses.

This represents a fundamental advance from "where the model looks" to "how spatial tissue organization influences outcome"—moving from correlation to potential biological mechanism.

4.3. Methodological Implications

Handcrafted vs. Learned Representations: Our results suggest that thoughtfully designed handcrafted features, grounded in established statistical principles, can capture biological signals orthogonal to learned representations. This supports a hybrid approach combining domain knowledge with deep learning.

5. Conclusion

We have introduced a novel framework for biological knowledge discovery in computational pathology through handcrafted spatial features derived from astrophysical statistical methods. By adapting the two-point correlation function with robust edge correction, we demonstrate that explicit spatial modeling can reveal interpretable biological insights while achieving competitive predictive performance.

Our approach moves beyond the limitations of attention-based interpretability to provide natural language descriptions of spatial tissue relationships. The discovery of specific spatial biomarkers—such as homogeneous tumor clustering predicting poor treatment response—demonstrates

the potential for computational methods to advance biological understanding rather than merely optimize predictive metrics.

This work establishes spatial interpretability as a new paradigm for computational pathology, offering a path toward AI systems that enhance biological discovery while maintaining the transparency necessary for clinical adoption. By providing a transparent view into the model's reasoning and directly linking its predictions to understandable morphological spatial relations, our work helps bridge the gap between machine learning and clinical insight. This "Cosmos to Clinic" approach offers a promising direction for the development of more accurate, robust, and trustworthy AI systems in medicine.

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